

Question 1: Firewall Zone

bash

```
sudo firewall-cmd --new-zone=ITISS --permanent
sudo firewall-cmd --reload
sudo firewall-cmd --zone=ITISS --add-interface=ens33 --permanent
sudo firewall-cmd --zone=ITISS --add-service=ssh --permanent
sudo firewall-cmd --zone=ITISS --add-port=30000-35000/tcp
--permanent
sudo firewall-cmd --reload
sudo firewall-cmd --zone=ITISS --list-all
```

Question 2: HTTP Access with SSH Block

bash

```
sudo apt install apache2 -y
sudo systemctl start apache2
sudo systemctl enable apache2
sudo firewall-cmd --new-zone=ITISS --permanent
sudo firewall-cmd --reload
sudo firewall-cmd --zone=ITISS --add-interface=ens33 --permanent
sudo firewall-cmd --zone=ITISS --add-service=http --permanent
sudo firewall-cmd --zone=ITISS --remove-service=ssh --permanent
sudo firewall-cmd --zone=ITISS --add-rich-rule='rule
family="ipv4" source address="192.168.1.100" service name="ssh"
reject' --permanent
sudo firewall-cmd --reload
sudo firewall-cmd --zone=ITISS --list-all
```

Question 3: Port Forwarding

bash

```
sudo sysctl -w net.ipv4.ip_forward=1
echo "net.ipv4.ip_forward=1" | sudo tee -a /etc/sysctl.conf
sudo firewall-cmd --zone=public
--add-forward-port=port=8000:proto=tcp:toport=80 --permanent
```

```
sudo firewall-cmd --zone=public --add-masquerade --permanent
sudo firewall-cmd --reload
sudo firewall-cmd --zone=public --list-all
```

Question 4: pfSense VM-Based Filtering

pfSense Web Interface Configuration:

Aliases:

```
Name: Indian_Govt
gov.in
nic.in
india.gov.in
```

```
Name: Indian_Banks
sbi.co.in
icicibank.com
hdfcbank.com
```

```
Name: VM2_Allowed
microsoft.com
redhat.com
wikipedia.org
google.com
```

```
oracle.com
```

Firewall Rules VM1:

```
Action: Pass, Interface: LAN1, Source: LAN1 net, Destination:
Indian_Govt, Port: HTTP/HTTPS
```

```
Action: Pass, Interface: LAN1, Source: LAN1 net, Destination:
Indian_Banks, Port: HTTP/HTTPS
```

```
Action: Pass, Interface: LAN1, Source: LAN1 net, Destination:
cdac.in, Port: HTTP/HTTPS
```

```
Action: Pass, Interface: LAN1, Source: LAN1 net, Destination:
google.com, Port: HTTP/HTTPS
```

```
Action: Block, Interface: LAN1, Source: LAN1 net, Destination:
Any
```

Firewall Rules VM2:

Action: Pass, Interface: LAN2, Source: LAN2 net, Destination: VM2_Allowed, Port: HTTP/HTTPS

Action: Block, Interface: LAN2, Source: LAN2 net, Destination: Any

Question 5: pfSense User-Based Filtering

Squid Installation:

System → Package Manager → Available Packages → Install Squid

Squid Configuration:

Services → Squid Proxy Server

Enable: Checked

Proxy Interface: LAN

Port: 3128

Authentication Method: Local

Users Created:

Username: John

Password: [password]

Allowed: Indian_Govt, Indian_Banks, Google, CDAC

Username: Peter

Password: [password]

Allowed: Microsoft, RedHat, Wikipedia, Google, Oracle

SquidGuard Rules:

Services → SquidGuard Proxy Filter

Target Categories: Create for each allowed domain group

ACL: Assign users to respective target categories

Question 6: Snort IDS

bash

```
sudo apt install snort -y
```

```
sudo snort -V
```

```
sudo nano /etc/snort/rules/local.rules
```

local.rules content:

```
alert tcp any any -> $HOME_NET any (msg:"NMAP Port Scan Detected"; flags:S; threshold:type both, track by_src, count 10, seconds 60; sid:1000001; rev:1;)
```

```
alert icmp any any -> $HOME_NET any (msg:"Large ICMP Packet - Possible DOS"; dsize:>1000; sid:1000002; rev:1;)
```

```
alert icmp any any -> $HOME_NET any (msg:"ICMP Flood Detected"; threshold:type both, track by_src, count 50, seconds 10; sid:1000003; rev:1;)
```

```
alert tcp $HOME_NET any -> any $HTTP_PORTS (msg:"HTTP Access from Internal Network"; sid:1000004; rev:1;)
```

```
bash
```

```
sudo snort -T -c /etc/snort/snort.conf
```

```
sudo snort -A console -c /etc/snort/snort.conf -i ens33
```

Question 7: Wazuh SIEM

Server:

```
bash
```

```
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo apt-key add -
```

```
echo "deb https://packages.wazuh.com/4.x/apt/ stable main" |
```

```
sudo tee /etc/apt/sources.list.d/wazuh.list
```

```
sudo apt update
```

```
sudo apt install wazuh-manager -y
```

```
sudo systemctl start wazuh-manager
```

```
sudo systemctl enable wazuh-manager
```

```
sudo apt install wazuh-dashboard -y
```

Client Ubuntu:

```
bash
```

```
wget
```

```
https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.0-1_amd64.deb
```

```
sudo dpkg -i wazuh-agent_4.7.0-1_amd64.deb
```

```
sudo nano /var/ossec/etc/ossec.conf
```

```
sudo systemctl start wazuh-agent
```

```
sudo systemctl enable wazuh-agent
```

Client Windows:

Download wazuh-agent-4.7.0-1.msi

Install with Wazuh Server IP

Start Wazuh service

Question 8: Wazuh + Snort Integration

```
bash
```

```
sudo nano /var/ossec/etc/ossec.conf
```

Add to ossec.conf:

```
xml
```

```
<localfile>  
  <log_format>snort-full</log_format>  
  <location>/var/log/snort/alert</location>  
</localfile>
```

```
bash
```

```
sudo systemctl restart wazuh-agent
```

```
sudo snort -A console -c /etc/snort/snort.conf -i ens33
```

Question 9: Wireshark

```
bash
```

```
ip addr show
```

```
sudo wireshark &
```

Capture Filter:

```
ip.addr == 192.168.1.50
```

Generate Traffic:

```
bash
```

```
ping 192.168.1.1
```

```
nslookup www.cdac.in
```

Question 10: tcpdump

```
bash
```

```
sudo tcpdump -D
```

```
sudo tcpdump -i ens33 -w stm-exam.pcap
```

```
ping 192.168.1.1
```

```
dig www.cdac.in
```

```
sudo tcpdump -r stm-exam.pcap -nn
```

Question 11: Nginx Reverse Proxy

VM1 & VM2:

```
bash
```

```
sudo apt install apache2 -y
```

```
echo "<h1>Exam Portal</h1>" | sudo tee /var/www/html/index.html
```

```
sudo systemctl restart apache2
```

VM3:

```
bash
```

```
sudo apt install nginx -y
```

```
sudo nano /etc/nginx/nginx.conf
```

nginx.conf:

```
nginx
```

```
http {  
    upstream exam_server {  
        server 192.168.1.51:80;
```

```

    }
    upstream result_server {
        server 192.168.1.52:80;
    }
    server {
        listen 80;
        location /exam {
            proxy_pass http://exam_server/;
        }
        location /result {
            proxy_pass http://result_server/;
        }
    }
}

bash
sudo nginx -t
sudo systemctl restart nginx

```

Question 12: Nginx Load Balancer

VM1 & VM2:

```

bash
echo "<h1>Server 1</h1>" | sudo tee /var/www/html/index.html
sudo systemctl restart apache2

```

VM3:

```

bash
sudo nano /etc/nginx/nginx.conf

```

nginx.conf:

```

nginx
http {
    upstream backend_servers {
        server 192.168.1.51 weight=6;
        server 192.168.1.52 weight=1;
    }
}

```

```
server {  
    listen 80;  
    location / {  
        proxy_pass http://backend_servers;  
    }  
}
```

bash

```
sudo nginx -t
```

```
sudo systemctl restart nginx
```

```
for i in {1..14}; do curl -s http://192.168.1.50/; done
```